INCH-POUND

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MILITARY SPECIFICATION
VIEWER, DRIVER'S, NIGHT VISION, AN/VVS-2(V)

This specification is approved for use by US Army Communications-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the Viewer, Driver's Night Vision AN/VVS-2(V).

1.2 Classification. Driver's Viewers covered by this specification shall be of the following types as specified:

Type I	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)1A (M60 application).
Type II	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)2A (XM1 application).
Type III	-	Viewer, Driver's, Night Vision, AN/VVS-2(V)3 (FVS application).

2. APPLICABLE DOCUMENTS

2.1 Government Documents.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ, USA Communications-Electronics Command, ATTN: AMSEL-ED-TM Fort Monmouth, NJ 07703-5023 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5855

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2.1.1 Specification, standards and handbooks. The following specifications, standards and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-P-11268	-	Parts, Materials, and Processes Used in Electronic Equipment
MIL-M-13231	-	Marking of Electronic Items
MIL-O-13830	-	Optical Components For Fire Control Instruments; General Specification Governing the Manufacture, Assembly, and Inspection of
MIL-F-14072	-	Finishes for Ground Electronic Equipment
MIL-I-49040	-	Image Intensifier Assembly, 25 Millimeter, Microchannel Inverter, MX-9644/UV
MIL-L-49147	-	Lens Assembly, Eyepiece, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-L-49148	-	Lens Assembly, Objective, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-E-49149	-	Electronic Control Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-E-49150	-	Entrance Housing Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-P-49151	-	Power Adapter Assembly, Viewer, Driver's Night Vision, AN/VVS-2(V)
MIL-M-49361	-	Magnifier, Tapered Fiber Optic, Viewer, Driver's Night Vision, AN/VVS-2(V)

STANDARDS

MILITARY

MIL-STD-108	-	Definitions of and Basic Requirements for Enclosures for Electric and Electronic Equipment
MIL-STD-150	-	Photographic Lenses
MIL-STD-252	-	Wired equipment, classification of Visual and Mechanical Defects for Equipment, Electronic, Wired, and Other Devices
MIL-STD-454	-	Standard General Requirements for Electronic Equipment

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MIL-STD-781	-	Reliability Testing for Engineering Development, Qualification, and Production
MIL-STD-810	-	Environmental Test Methods
DOD-STD-2000A/4	-	General Purpose Soldering Requirements for Electrical and Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, 700 Robbins Avenue, Building #4, Section D, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings and publications. The following other Government documents, drawings and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

USA COMMUNICATIONS-ELECTRONICS COMMAND

SM-D-771480-1	-	Viewer, Driver's Night Vision AN/VVS-2(V)1A
SM-D-771480-2	-	Viewer, Driver's Night Vision AN/VVS-2(V)2A
SM-D-771480-3	-	Viewer, Driver's Night Vision AN/VVS-2(V)3
SM-D-771335	-	Housing Assembly - Main Viewer
SM-D-771212	-	Tube and Magnifier Assembly, Image Intensifier

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

2.3 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM B26	-	Aluminum Alloy Sand Castings
ASTM B108	-	Aluminum Alloy Permanent Mold Castings

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

3. REQUIREMENTS

3.1 Description. The Viewer, Driver's Night Vision, AN/VVS-2(V), herein referred to as the viewer, is a biocular display viewer used for closed hatch driving of combat vehicles. The viewer consists of an objective lens assembly, fiber optic magnifier, eyepiece lens assembly, entrance window and mirror housing assembly, main viewer housing assembly which includes an electronic control, assembly, an electric power adapter, and image intensifier tube which is part of the tube and magnifier assembly.

3.2 Construction. The viewer shall be constructed as specified herein and in accordance with the following drawings as applicable:

Type I - SM-D-771480-1

Type II - SM-D-771480-2

Type III - SM-D-771480-3

3.3 Qualification. Unless otherwise specified by the contract or purchase order, viewers furnished under this specification shall be products approved as qualified products (see 3.13.2 and 6.3).

3.3.1 Initial production testing (IPT). When specified in the contract (see 6.2.b) the contractor shall furnish unit(s) for IPT inspection and approval (see 4.4).

3.4 Parts and materials. Parts and materials shall be as specified herein and as shown on applicable drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification and shall conform to MIL-P-11268.

3.5 Major components. The following major components shall be constructed and tested in accordance with applicable drawings, specifications and as specified herein.

3.5.1 Objective lens assembly. The objective lens assembly shall conform to MIL-L-49148.

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3.5.2 Fiber optic magnifier. The fiber optic magnifier shall conform to MIL-M-49361.

3.5.3 Power adapter assembly. The power adapter assembly shall conform to MIL-P-49151.

3.5.4 Entrance window and mirror housing assembly. The entrance window and mirror housing assembly shall conform to MIL-E-49150, Type I and Type II as applicable.

3.5.5 Eyepiece lens assembly. The eyepiece lens assembly shall conform to MIL-L-49147.

3.5.6 Image intensifier. The image intensifier (MX-9644/UV) shall conform to MIL-I-49040.

3.5.7 Tube and magnifier assembly. The tube and magnifier assembly shall conform to SM-D-771212 and MIL-M-49361.

3.5.8 Main viewer housing assembly.

3.5.8.1 Construction. The main viewer housing assembly shall be constructed in accordance with applicable drawings of SM-D-771335. The electric power adapter assembly and electronic control assembly are part of this assembly.

3.5.9 Electronic control assembly. The electronic control assembly shall conform to MIL-E-49149.

3.6 Performance characteristics.

3.6.1 ON/OFF/Gain switch. The switch shall turn the viewer ON and OFF, and vary the brightness of the display.

3.6.2 Viewer bearing rotation. The viewer shall rotate 45 degrees in either direction from the center detent position for Type I, and 30 degrees in either direction from the center detent position for Type II and Type III. Torque required shall be between 3 and 6 foot pounds, excluding the detent position.

3.6.3 Electronic control. When installed in the viewer the electronic control shall turn off the viewer when the ambient light level exceeds 4 footlamberts.

3.6.4 Field of view. The viewer's field of view shall have a minimum width of 44 degrees horizontal and 35 degrees vertical.

3.6.5 Resolution. The viewer shall have a minimum on-axis resolution of 0.8 line pairs per milliradian.

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3.6.6 Magnification. The on-axis magnification of the viewer shall be 0.93, +0.09/-0.06 measured at 50 millimeter (mm) eye relief.

3.6.7 Linear distortion. The linear distortion of the viewer measured from an on-axis pupil at 50 mm eye relief shall not exceed 13 percent.

3.6.8 Objective focus. The objective shall be fixed focus for an object distance of 50, ± 5 feet.

3.6.9 Eyepiece focus. The eyepiece focus when measured from 50 mm eye relief shall be set at -2.0, +0.3/-0.1 diopters on axis.

3.7 Environmental characteristics.

3.7.1 High temperature.

3.7.1.1 Operation. The viewer shall operate at a temperature up to +52°C without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.1.2 Storage. The viewer shall withstand storage at a temperature of +68°C without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.2 Low temperature.

3.7.2.1 Operation. The viewer shall operate at a temperature down to -51°C without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.2.2 Storage. The viewer shall withstand storage at a temperature of -51°C without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.3 Temperature shock. The viewer shall withstand cycles of sudden changes in temperature between -51°C and +68°C without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.4 Humidity. The viewer shall withstand +68°C temperature and 94, ± 4 percent humidity without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.5 High pressure water stream. The viewer and associated equipment shall be operable and undamaged during and after exposure to a high pressure stream of water without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.6 Sealing. The viewer shall be filled with a mixture of dry nitrogen with 1 mole-percent helium and shall be sealed. The helium leak rate of the sealed viewer shall not be greater than 9.9×10^{-4} cm^3/sec. The viewer shall withstand sealing test without damage or degradation of performance (see 3.13.1 and 3.13.6).

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3.7.7 Vibration. The viewer shall withstand vibration equivalent to that encountered in tracked vehicles, during 5000 miles of operation without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.8 Shock.

3.7.8.1 Basic shock. The viewer shall withstand a basic shock pulse of 40 g's (see 3.13.4) with a pulsewidth of 18 milliseconds without damage or degradation of performance (see 3.13.1 and 3.13.6).

3.7.8.2 Weapon shock. The viewer shall withstand simulated weapon firing shock pulse of 100 g's with a pulsewidth of 1.5 millisecond one half sine pulse without damage and degradation of performance (see 3.13.1 and 3.13.6).

3.7.9 Fungus. The viewer shall not support fungus growth nor be damaged (see 3.13.1) by a fungus atmosphere or adjoining fungus growth.

3.7.10 Salt fog. The viewer shall not be damaged (see 3.13.1) by exposure to salt fog.

3.7.11 Sand and Dust. The viewer with glass lenses covered shall not be damaged (see 3.13.1) by the environment of blowing sand and dust at 35 knots.

3.7.12 Environmental stress screening (ESS). Each assembly shall be exposed to environmental stress screening, designed to surface infant mortality type failures, as established in accordance with Task 401 of MIL-STD-781 by the contractor and approved by the Government.

3.8 Reliability. The mean time to failure (MTTF) of the viewer shall be 1000 hours minimum.

3.9 Interchangeability. The entrance window and mirror housing assembly, the objective lens assembly, and the eyepiece lens assembly shall be physically interchangeable between viewer assemblies without degradation of the resolution (see 3.6.5), objective focus (see 3.6.8), and eyepiece focus (see 3.6.9) requirements. These interchanges shall be made without any modification or adjustment, beyond that necessary during replacement, of any of the parts or assemblies involved.

3.10 Treatment and painting. Unless otherwise specified, treatment and painting shall conform to MIL-F-14072.

3.10.1 Marking. Unless otherwise specified, the viewer and all components shall be marked and identified per MIL-M-13231.

3.11 Castings. Unless otherwise specified, castings shall conform to ASTM B26 or ASTM B108.

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3.11.1 Welding. Unless otherwise specified, welding shall conform to Requirement 13 of MIL-STD-454.

3.12 Workmanship. Standards of workmanship shall be such that components will meet all requirements of this specification and any referenced specifications or drawings where inspected in accordance with section 4. Parts and assemblies shall exhibit none of the defects of type 9 or type 11 as categorized in MIL-STD-252. Applicable parts of Requirement 9 of MIL-STD-454 apply. Soldering shall conform to the requirements of DOD-STD-2000A/4.

3.12.1 Cleanliness. The objective lens assembly, the eyepiece, and the entrance window housing assembly shall meet the cleanliness requirement for optical systems as specified in MIL-O-13830.

3.13 Technical interpretations. The following technical interpretations are, when referenced in sections 3, 4, or 5, mandatory for this specification.

3.13.1 Damage. Damage is defined as the following:

a. Breakage, loosening, shift, dents, failure of finish, hardware, connection or component.

b. Condensation, fogging or moisture on any internal surface.

c. Failure to operate or operate properly.

3.13.2 Qualified product. A product which has successfully met all of the requirements of an IPT or first article witnessed by the CECOM, Concurrent Engineering Directorate or its designated representative. Should the supplier make substantial changes in the process, personnel, location, or equipment used to produce a qualified product, or have not delivered the qualified product to the Federal Government within 12 months of contract award, an IPT shall be performed to the extent necessary to verify continued compliance to specification requirements for all characteristics of the product affected by the changes.

3.13.3 Chargeable failure. A chargeable system failure during the reliability test of 4.8 is the inability to pass the tests of 4.6.1, 4.6.2, 4.6.3, or 4.6.5.

3.13.4 "g". "g" is defined as an acceleration or deceleration of 32.17 feet per second per second.

3.13.5 Ambient temperature. Ambient temperature is defined as +23°C, ± 10°C.

3.13.6 Degradation of performance. Any deterioration in quality, or failure to operate in accordance with the requirements of this specification.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspection. Inspection shall be classified as follows:

a. Inprocess inspection (4.3).
b. Initial production testing (4.4).
c. Quality conformance inspection (4.5).
d. Packaging inspection (4.10)

4.3 Inprocess inspection. Prior to assembly, each component listed in Table I shall be inspected in accordance with its requirements as specified below.

4.3.1 Inprocess inspection prescribed by a specification or drawing. All materials, parts, processes and assemblies shall be inspected for conformance to the requirements of Table I.

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TABLE I. Inprocess inspection.

Parts, Processes or Assemblies	Requirement paragraph	Acceptance criteria
Parts & materials	3.4	MIL-P-11268
Treatment & painting	3.10	MIL-F-14072
Marking	3.10.1	MIL-M-13231
Workmanship	3.12	MIL-STD-252, MIL-STD-454
Castings	3.11	ASTM B 26, ASTM B108
Cleanliness	3.12.1	MIL-O-13830
Objective lens assembly	3.5.1	MIL-L-49148
Fiber optic magnifier	3.5.2	MIL-M-49361
Power adapter assembly	3.5.3	MIL-P-49151
Entrance assembly	3.5.4	MIL-E-49150
Eyepiece assembly	3.5.5	MIL-L-49147
Image intensifier	3.5.6	MIL-I-49040
Tube and magnifier assembly	3.5.7	SM-D-771212
Main viewer housing assembly	3.5.8	SM-D-771335
Electronic control assembly	3.5.9	MIL-E-49149

4.3.2 Environmental stress screening(ESS). Each assembly shall be subjected to ESS in accordance with Task 401 of MIL-STD-781. Sufficient stress must be applied to minimize infant mortality, in the field.

4.4 Initial production testing. Unless otherwise specified in the contract or purchase order, the initial production testing shall be performed by the contractor (see 6.2.b).

4.4.1 Unit tests. Using assemblies that have passed the examination of 4.3.1 and ESS of 4.3.2, the IPT units shall be subjected to the tests in Table II. Prior to the start of the tests, the viewers shall be identified as to their applicable subgroups. Tests shall be conducted in the order listed. Failure of any test shall constitute failure of the IPT.

4.4.2 IPT testing sequence. Unless otherwise specified (see 6.2.b), the IPT testing shall consist of ten (10) viewers. Three (3) shall be used in Subgroup 1, three (3) shall be used in subgroup 2 and four (4) shall be used in Subgroup 3. In Subgroup 2 one (1) each viewer shall be used for salt fog, sand and dust, and fungus.

4.4.3 Disposition of IPT samples. Disposition of IPT samples shall be as specified in the contract or purchase order (see 6.2.b).

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TABLE II. Initial production testing.

Inspection	Requirement paragraph	Test paragraph	Subgroups 1	2	3
ON/OFF/gain switch	3.6.1	4.6.1	X	X	X
Viewer bearing rotation	3.6.2	4.6.2	X	X	X
Electronic control	3.6.3	4.6.3	X	X	X
Field of view	3.6.4	4.6.4	X	X	X
Resolution	3.6.5	4.6.5	X	X	X
Magnification	3.6.6	4.6.6	X	X	X
Linear distortion	3.6.7	4.6.7	X	X	X
Objective focus	3.6.8	4.6.8	X	X	X
Eyepiece focus	3.6.9	4.6.9	X	X	X
High temperature	3.7.1	4.7.1	X		
Low temperature	3.7.2	4.7.2	X		
Temperature shock	3.7.3	4.7.3	X		
Humidity	3.7.4	4.7.4	X		
High pressure water stream	3.7.5	4.7.5	X		
Vibration (4.7.7a only)	3.7.7	4.7.7a	X		
Shock	3.7.8	4.7.8	X		
Sealing	3.7.6	4.7.6	X	X	X
Fungus	3.7.9	4.7.9		X	
Salt fog	3.7.10	4.7.10		X	
Sand and dust	3.7.11	4.7.11		X	
Reliability	3.8	4.8			X
Interchangeability	3.9	4.9	X	X	X

4.5 Quality conformance inspection.

4.5.1 Group A inspection. Each viewer shall be examined in accordance with the requirements of 4.3 and shall be tested in accordance with Table III in the order listed. Failure of any test shall be cause for rejection of that unit. Viewers which pass Group A inspection shall be formed into lots by the contractor.

TABLE III. Group A inspection.

Inspection	Requirement paragraph	Test paragraph
ON/OFF gain switch	3.6.1	4.6.1
Viewer bearing rotation	3.6.2	4.6.2
Electronic control	3.6.3	4.6.3
Resolution	3.6.5	4.6.5
Objective focus	3.6.8	4.6.8
Eyepiece focus	3.6.9	4.6.9

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4.5.2 Group B inspection. Group B inspection shall be conducted on lots which have passed the tests in 4.5.1. Sampling of lots shall be in accordance with the requirements of the contract or purchase order (see 6.2.c). Tests shall be conducted in accordance with Table IV.

TABLE IV. Group B inspection.

Inspection	Requirement paragraph	Test paragraph
Field of view	3.6.4	4.6.4
Magnification	3.6.6	4.6.6
Linear distortion	3.6.7	4.6.7

4.5.3 Group C inspection. Group C inspections shall be conducted on lots which have passed the Group A tests. The sample(s) shall be subjected to the inspections specified in Table V. Samples shall be selected as specified in the contract or purchase order (see 6.2.c). Tests shall be performed in the order listed.

TABLE V. Group C inspection.

Inspection	Requirement paragraph	Test paragraph
Subgroup 1	3.7.7	4.7.7b
Vibration (4.7.7b)	3.7.1	4.7.1
High temperature	3.7.2	4.7.2
Low temperature	3.7.6	4.7.6
Sealing		
Subgroup 2		
Temperature shock	3.7.3	4.7.3
High pressure water stream	3.7.5	4.7.5
Vibration (4.7.7a)	3.7.7	4.7.7a
Shock	3.7.8	4.7.8
Subgroup 3		
Humidity	3.7.4	4.7.4

4.5.3.1 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or order (see 6.2.d).

4.5.4 Group D inspection. Group D inspections shall be conducted on viewers selected from units which have passed the tests in 4.5.1. Tests shall be conducted in accordance with Table VI.

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4.5.4.1 Sampling for Group D inspection. The reliability acceptance test shall be performed on viewers from the lot submitted in every sixth month of production. The sample size shall be 4 viewers.

TABLE VI. Group D inspection.

Inspection	Requirement paragraph	Test paragraph
Reliability	3.8	4.8

4.5.4.2 Group D failures. Actions required relative to Group D failures shall be as specified in the contract or purchase order (see 6.2.d).

4.5.4.3 Reinspection of Group C and D sample units. Unless otherwise specified, sample units which have been subject to and passed Group C and D inspection may be accepted on contract or purchase order provided all damage is repaired and the units are resubjected to and pass Group A inspection.

4.6 Operational tests. Viewer failures attributable to image intensifiers failure to meet the requirements of MIL-I-49040 shall not be counted against the viewer. After image intensifier failure, the intensifier shall be replaced and the test resumed with time commencing at the point of failure. Only salt fog and sand and dust shall be performed with the protective covers installed. The operational tests required by MIL-STD-810 shall be performed per 4.6.1, 4.6.2, 4.6.3, and 4.6.5 unless otherwise specified. Unless otherwise specified, an external power supply furnishing 28, ± 2Vdc to the power adapter assembly shall be used for all tests in lieu of batteries. Unless otherwise specified, all tests shall be performed at ambient temperature (see 3.13.5). Temperature chambers shall maintain specified temperatures within ± 2.0°C.

4.6.1 ON/OFF/Gain switch. Operate the switch, using first the power adapter and then the battery to provide power. Remove battery before applying vehicle power. Failure to meet the requirements of 3.6.1 shall constitute failure of this test.

4.6.2 Viewer bearing rotation. Rotate the viewer by means of a torque wrench. Failure to meet the requirements of 3.6.2 shall constitute failure of this test.

4.6.3 Electronic control. Turn the viewer ON and observe a target area whose average brightness is less than one footlambert. Raise the target brightness to 4 footlamberts. Failure to meet the requirements of 3.6.3 shall constitute failure of this test.

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4.6.4 Field of view. Observe a target point imaged at the extreme left edge of the viewer display. The point shall be visible at the right edge after the viewer is rotated horizontally through the angle specified in 3.6.4 Perform a similar procedure for the vertical field of view. Failure to meet the requirements of 3.6.4 shall constitute failure of this test.

4.6.5 Resolution. Place a high contrast (per MIL-STD-150) USAF 1951 Resolving Power Test Target (clear lines on black background) with a collimator focused to give an apparent image distance of 50 feet from the driver's viewer system. Back illuminate the target with 2854 K, ± 200 K light of such intensity that the brightness of the target through the collimator can be set at a maximum of 10^{-3} footlamberts. View the test target binocularly without any visual aids and determine the high light level resolution. Failure to meet the requirements of 3.6.5 shall constitute failure of this test.

4.6.6 Magnification. This test shall be performed using the test method of 4.6.9, and may be performed concurrently with the eyepiece focus test. Failure to meet the requirement of 3.6.6 shall constitute failure of this test.

4.6.7 Linear Distortion. This test shall be performed using the test method of 4.6.9, and may be performed concurrently with the eyepiece focus test. Failure to meet the requirement of 3.6.7 shall constitute failure of this test.

4.6.8 Objective focus. Set three USAF 1951 resolving power test targets (or equivalent) at 40, 50 and 70 feet from the viewer. With a diopterscope, observe the smallest pattern at each distance resolvable through the viewer. Calculate the angular resolution at each target. The target at 50 feet shall have the smallest angular resolution. Failure to meet the requirement of 3.6.8 shall constitute failure of this test.

4.6.9 Eyepiece focus. The viewer shall be placed on a test fixture as depicted in Figure 1. The rotary axis of the viewer is perpendicular to the input line of sight. Place a diopterscope with its optical axis coincident with the system output axis and its front aperture at an eye relief of 50 mm. The center of the diopterscope rotary table shall also be at a 50 mm eye relief. Measure the diopter focus at the field of view. Rotate the system table right and left 22 degrees from the center and measure the rotation (angles) of the diopterscope table required to see the target under these conditions. Next rotate the system table left and right 3 degrees (6 degrees total) and record the total rotation of the diopter table to see the target. Magnification for the viewer is M and distortion is D. Failure to meet requirements of 3.6.9 shall constitute failure of this test.

$$M = \frac{\tan \Theta}{\tan 6^\circ}$$

Θ, = angle subtended by diopterscope table for a system rotation of 6°

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D = 100 x ((Tan ϕ_2/M Tan 22)-1)

ϕ_2 = Average of right and left rotation of diopterscope for 22° system rotation

4.7 Environmental tests. All requirements specified in 4.6 apply during Environmental Testing.

4.7.1 High temperature. The viewer shall be subjected to the test of Method 501.1, Procedure I, MIL-STD-810, except that the highest operating temperature of Steps 4 and 5 shall be +52°C. The highest storage temperature of Steps 2 and 3 shall be +68°C. Steps 4 and 6 stabilization period shall be a minimum of two hours. Step 5 operating test shall be the ON/OFF/gain switch test of 4.6.1, performed using the power adapter only. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, and 3.7.1 shall constitute failure of this test.

4.7.2 Low temperature. The viewer shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -51°C and shall be maintained until temperature stabilization of the test item is reached. Steps 2, 4, and 6 stabilization period shall be a minimum of two hours. Step 5 operating test shall be the ON/OFF/gain switch test of 4.6.1, performed using the power adapter only. Low operating temperature (Step 4) shall be -51°C. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, and 3.7.2 shall constitute failure of this test.

4.7.3 Temperature shock. The viewer shall be subjected to the test of Method 503.1, Procedure I of MIL-STD-810, except the high temperature chamber shall be maintained at +68 degrees C, the low temperature chamber shall be maintained at -51 degrees C, and Steps 1, 3, and 5 stabilization period shall be a minimum of two hours. Immediately after each cold cycle, the viewer shall be examined for internal condensation. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, or 3.7.3 shall constitute failure of this test. Evidence of moisture or condensation on any internal surface or failure to meet the requirements of Method 503.1, Step 9 shall constitute failure of this test.

4.7.4 Humidity. The viewer shall be subjected to the test of Method 507.1, Procedure II, of MIL-STD-810 except maximum temperature shall be +68°C. Step 9 shall consist of requirements 3.6.1, 3.6.2, 3.6.3, 3.6.5 or 3.7.4 of this specification. Failure of meet these requirements shall constitute failure of this test.

4.7.5 High pressure water stream. The viewer shall be subjected to the test of MIL-STD-108, Table II, watertight. During the 60-minute test period, the viewer shall be rotated in its mount through the angles specified in 3.6.2. Only the portion of the viewer exterior to the vehicle shall be tested. Failure to meet the requirement of 3.6.1, 3.6.2, 3.6.3, 3.6.5 or 3.7.5 specification shall constitute failure of this test.

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4.7.6 Sealing. The viewer shall be subjected to the test of Method 512.1, Procedure IV of MIL-STD-810. The viewer shall be tested for helium leakage with a Varian, Model 925-40 "Porta-Test" or equivalent. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, and 3.7.6 shall constitute failure of this test.

4.7.7 Vibration.

a. The viewer while operating shall be subjected to the test of MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6, for tracked vehicles.

b. The viewer shall be subjected to MIL-STD-810, Method 514.2, Procedure VIII, Table 514.2-6 tracked vehicles, limited to one sweep in the vertical axis only.

Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, and 3.7.7 shall constitute failure of this test.

4.7.8 Shock. The viewer shall be subjected to the test of MIL-STD-810, Method 516.2.

a. 40 g's at 18 milliseconds, Figure 516.2-1 shall be the basic design shock Procedure I.
b. Weapon firings, 100 g's using Procedure IV except at 1.5 milliseconds (ms), Figure 516.2-2.

Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5, and 3.7.8 shall constitute failure of this test.

4.7.9 Fungus. The viewer shall be subjected to the Test Method 508.1, Procedure I of MIL-STD-810. At the conclusion of the exposure period, examination shall reveal no evidence of visible fungus or corrosion on any equipment surfaces. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5 or 3.7.9 shall constitute failure of this test.

4.7.10 Salt fog. The viewer with protective covers installed shall be subjected to the test of Method 509.1, Procedure I of MIL-STD-810. Failure to meet the requirements of 3.6.1, 3.6.2, 3.6.3, 3.6.5 or 3.7.10 shall constitute failure of this test.

4.7.11 Sand and dust. The viewer with protective covers installed shall be subjected to the test Method 510.1, MIL-STD-810. Failure to meet the requirement of 3.6.1, 3.6.2, 3.6.3, 3.6.5 or 3.7.11 shall constitute failure of this test.

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4.8 Reliability. The viewer reliability requirement shall be demonstrated by a durability test on optical/mechanical components. A test accept decision shall be based on demonstration of 1000 hours accumulated on each viewer with no chargeable viewer failures (see 3.13.3). A test reject decision shall be based on occurrence of one chargeable failure in 1000 hours for any viewer. Any tube failures shall be repaired or replaced and the viewer put back on test with time accumulated prior to the failure countable. The number of units under test shall be four. Failure to meet the requirements of 3.8 shall constitute failure of this test.

4.8.1 Reliability test conditions. The reliability test shall be conducted in accordance with the following test profile:

a. The test conditions shall be:

Temperature	0°C 50% of test time 30°C 50% of test time
Vibration	2.5 g's at fixed non-resonant frequencies between 10 and 50 Hz
Shock	100 g's at 1.5 ms, 1 shock/hour of operation.
Performance test	The viewers shall be tested every 160 hours ± 25 hours for failure. Test per 4.6.1, 4.6.2, 4.6.3, and 4.6.5.
Light levels	The light level shall be 1 x 10^{-4} footlambert on scene increased to 100 footlamberts for 10 seconds minimum at least once during each 25 hours of operation. Instrumentation sufficient to insure immediate recognition of catastrophic failure shall be provided.

b. Viewer in storage position non-operating with shock and vibration applied 75 percent of the time.

c. Viewer in operating position in operation with shock and vibration 20 percent of the time.

d. Viewer statically operating 5 percent of the time.

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4.9 Interchangeability. All IPT assemblies which have passed the resolution (see 3.6.5), objective focus (see 3.6.8), and eyepiece focus (see 3.6.9) requirements shall be tested for interchangeability. For this test, the mount assemblies, entrance window and mirror housing assemblies, objective lens assemblies, and eyepiece lens assemblies shall be interchanged between viewer assemblies as shown in Figure 2 and the resolution (see 4.6.5), objective focus (see 4.6.8), and eyepiece focus (see 4.6.9) tests repeated after each test arrangement and after reassembly to the pretest arrangement. This test shall be repeated for all additional IPT assemblies until each IPT assembly has been tested once as shown in Figure 2 with another IPT assembly. Failure to meet the requirements of 3.6.5, 3.6.8, 3.6.9, and 3.9 after each arrangement shall constitute failure of this test.

4.10 Packaging inspection. The inspection of the preservation-packaging packing and marking shall be in accordance with the requirements of the contract or purchase order.

5. PACKAGING

5.1 Packaging requirements. The packaging requirements for the desired level(s) of protection shall be as specified by the acquisition activity (see 6.2.i).

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The viewer is intended for closed hatch driving of combat vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification.

b. Quantity and schedule of initial production testing and disposition of IPT samples.

c. Sampling plans for Group B and Group C testing. As guidance, unless otherwise specified, sampling shall be conducted per the requirements of MIL-STD-105.

d. Actions required relative to Group C and Group D failures,(see 4.5.3.1 and 4.5.4.2).

e. A prohibition against delivering viewers with batteries installed.

f. That the contractor will furnish necessary batteries, of all types specified, for inspection and test.

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g. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.1.2).

h. MIL-STD-810C is used for environmental tests where MIL-STD-810 is specified.

i. Packaging requirements.

6.3 Qualification. With respect to products requiring qualification, unless otherwise specified by the contract or purchase order, if a product is not qualified at time of award, the contract shall require qualification prior to first delivery. The activity responsible for approving qualified products is the CECOM, Concurrent Engineering Directorate, ATTN: AMSEL-ED-IN, Fort Monmouth, NJ 07703. Information pertaining to qualification of products may be obtained from the above activity.

6.4 Definitions. see 3.13.

6.5 International standardization agreement. Provisions of this publication are the subject of an international standardization agreement (QSTAG 579). When amendment, revision, or cancellation of this publication is proposed that will affect or violate the agreement concerned, the preparing activity will take proper action through international standardization channels. If local office is not available contact Commander, US Army Material Development & Readiness Command, ATTN: DRCIRD, 5001 Eisenhower Avenue, Alexandria VA 22333.

6.6 Subject term keyword listing.
Driver's Viewer

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Army-CR

Preparing Activity:
Army-CR

Project: 5855-A358

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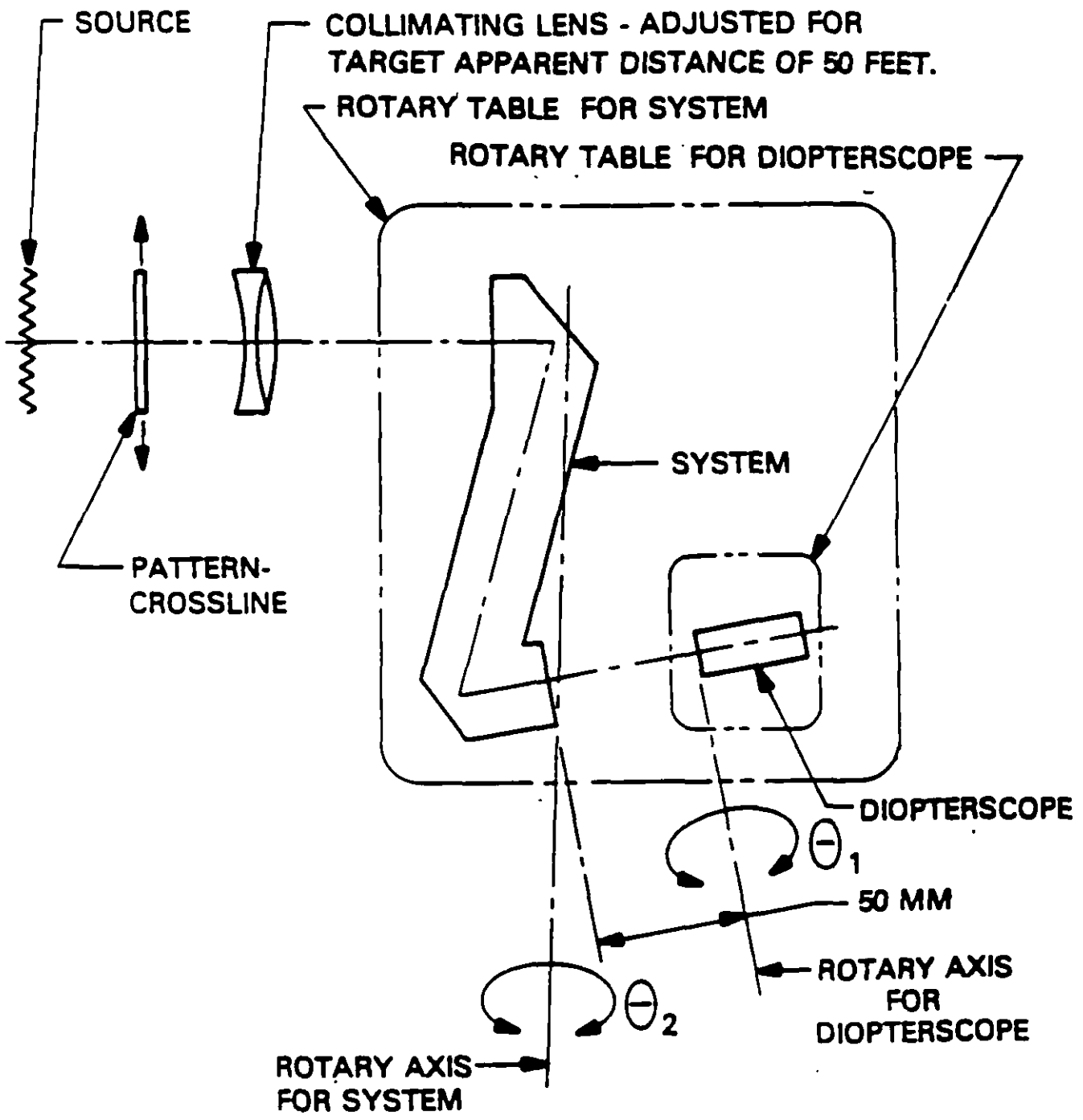


FIGURE 1. SYSTEM TESTS DIAGRAM

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PRETEST ARRANGEMENT

VIEWER 1	VIEWER 2
EW&MHA-1	EW&MHA-2
OLA-1	OLA-2
MA-1	MA-2
ELA-1	ELA-2

TEST ARRANGEMENT 1

VIEWER 1	VIEWER 2
EW&MHA-2	EW&MHA-1
OLA-1	OLA-2
MA-1	MA-2
ELA-1	ELA-2

TEST ARRANGEMENT 2

VIEWER 1	VIEWER 2
EW&MHA-2	EW&MHA-1
OLA-2	OLA-1
MA-1	MA-2
ELA-1	ELA-2

TEST ARRANGEMENT 3

VIEWER 1	VIEWER 2
EW&MHA-2	EW&MHA-1
OLA-2	OLA-1
MA-1	MA-2
ELA-2	ELA-1

TEST ARRANGEMENT 4

VIEWER 1	VIEWER 2
EW&MHA-1	EW&MHA-2
OLA-2	OLA-1
MA-1	MA-2
ELA-2	ELA-1

TEST ARRANGEMENT 5

VIEWER 1	VIEWER 2
EW&MHA-1	EW&MHA-2
OLA-1	OLA-2
MA-1	MA-2
ELA-2	ELA-1

EW&MHA - Entrance Window and Mirror Housing Assembly
OLA - Objective Lens Assembly
MA - Mount Assembly
ELA - Eyepiece Lens Assembly

FIGURE 2. INTERCHANGEABILITY INSPECTION ARRANGEMENTS

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-V-49082D(CR)	22 May 1992

3. DOCUMENT TITLE
VIEWER, DRIVER'S NIGHT VISION, AN/VVS-2(V)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
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